Microsoft Elevate

# CAPSTONE PROJECT

# AIRLINE CUSTOMER BEHAVIOR ANALYTICS DASHBOARD USING POWER BI


**PRESENTED BY**

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# OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

# PROBLEM STATEMENT:

Airlines handle a large number of passengers every day, making it difficult to understand customer behavior and satisfaction levels. Passenger experience is influenced by several factors such as travel class, service quality, and flight delays.

Although airlines collect a huge amount of passenger data, it is not easy to extract meaningful insights from the raw data without proper analytical tools. Understanding what affects passenger satisfaction is important for improving airline services and enhancing the overall travel experience.

Therefore, there is a need for a system that can analyze passenger data and provide clear insights into customer behavior and service performance

# PROPOSED SOLUTION:

Airlines handle a large number of passengers every day, making it difficult to understand customer behavior and satisfaction levels. Passenger experience is influenced by several factors such as travel class, service quality, and flight delays.

Although airlines collect a huge amount of passenger data, it is not easy to extract meaningful insights from the raw data without proper analytical tools. Understanding what affects passenger satisfaction is important for improving airline services and enhancing the overall travel experience.

Therefore, there is a need for a system that can analyze passenger data and provide clear insights into customer behavior and service performance

# SYSTEM APPROACH:

The project was developed using **Power BI**, which is a powerful tool for data analysis and visualization.

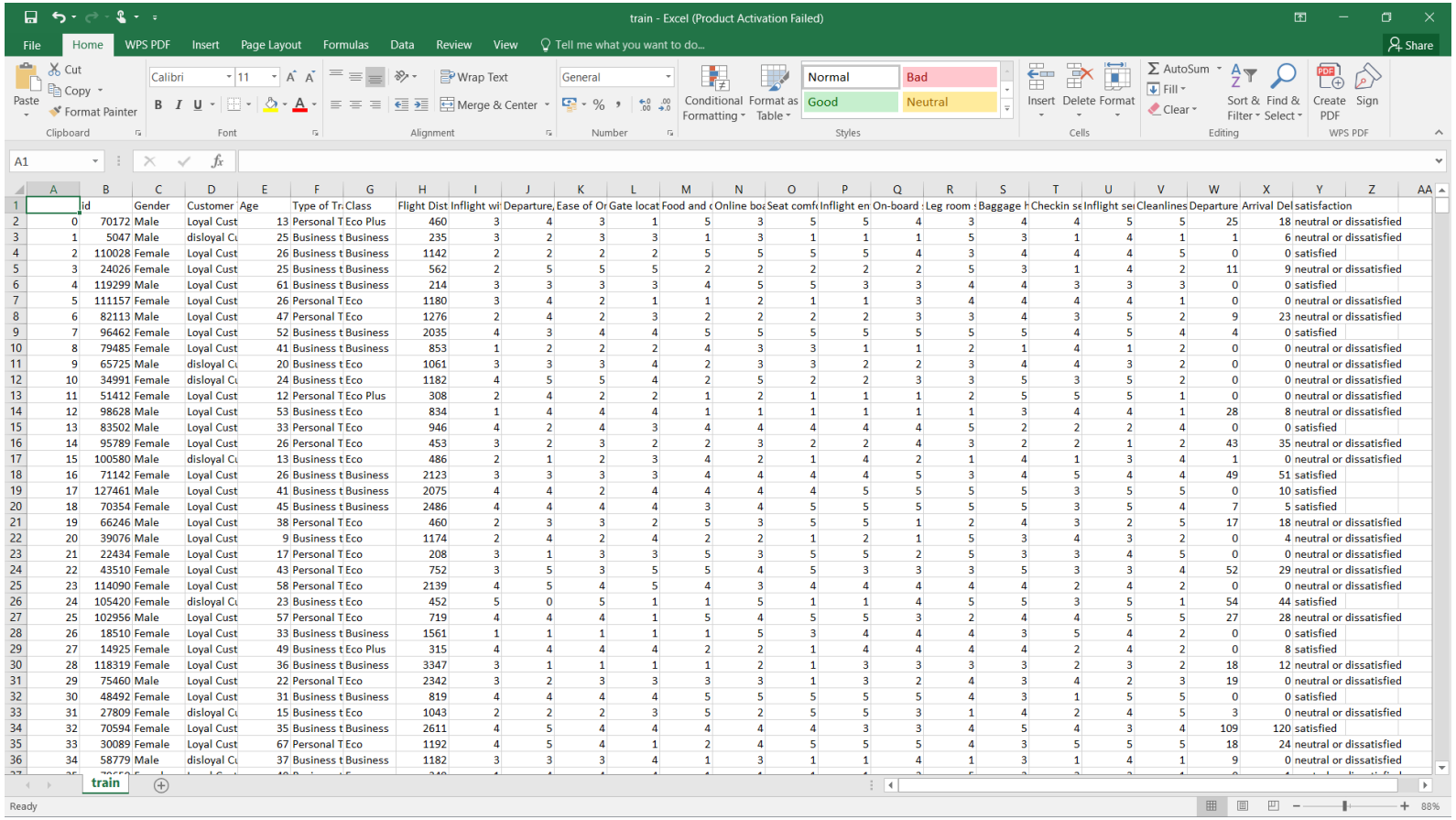
The development process included the following steps:

- Collecting the airline passenger dataset.
- Cleaning and preparing the dataset using Power Query.
- Creating additional columns to better analyze passenger behavior.
- Developing analytical measures using DAX.
- Designing an interactive dashboard with multiple visualizations.

**Tools Used**

- Power BI Desktop – Dashboard creation and data visualization
- Power Query – Data cleaning and transformation
- DAX – Creating calculated measures and metrics

# SYSTEM APPROACH:



| | id | Gender | Customer | Age | Type of Tr | Class | Flight Dist | Inflight wi | Departure | Ease of Or | Gate locat | Food and | Online bo | Seat comf | Inflight en | On-board | Leg room | Baggage h | Checkin se | Inflight se | Cleanlines | Departure | Arrival Del | satisfaction |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | | | | | | |
| 2 | 0 | 70172 | Male | Loyal Cust | 13 | Personal T | Eco Plus | 460 | 3 | 4 | 3 | 1 | 5 | 3 | 5 | 5 | 4 | 3 | 4 | 4 | 5 | 5 | 25 | 18 | neutral or dissatisfied |
| 3 | 1 | 5047 | Male | disloyal Cu | 25 | Business t | Business | 235 | 3 | 2 | 3 | 3 | 1 | 3 | 1 | 1 | 1 | 5 | 3 | 1 | 4 | 1 | 1 | 6 | neutral or dissatisfied |
| 4 | 2 | 110028 | Female | Loyal Cust | 26 | Business t | Business | 1142 | 2 | 2 | 2 | 2 | 5 | 5 | 5 | 5 | 4 | 3 | 4 | 4 | 4 | 5 | 0 | 0 | satisfied |
| 5 | 3 | 24026 | Female | Loyal Cust | 25 | Business t | Business | 562 | 2 | 5 | 5 | 5 | 2 | 2 | 2 | 2 | 2 | 5 | 3 | 1 | 4 | 2 | 11 | 9 | neutral or dissatisfied |
| 6 | 4 | 119299 | Male | Loyal Cust | 61 | Business t | Business | 214 | 3 | 3 | 3 | 3 | 4 | 5 | 5 | 3 | 3 | 4 | 4 | 3 | 3 | 3 | 0 | 0 | satisfied |
| 7 | 5 | 111157 | Female | Loyal Cust | 26 | Personal T | Eco | 1180 | 3 | 4 | 2 | 1 | 1 | 2 | 1 | 1 | 3 | 4 | 4 | 4 | 4 | 1 | 0 | 0 | neutral or dissatisfied |
| 8 | 6 | 82113 | Male | Loyal Cust | 47 | Personal T | Eco | 1276 | 2 | 4 | 2 | 3 | 2 | 2 | 2 | 2 | 3 | 3 | 4 | 3 | 5 | 2 | 9 | 23 | neutral or dissatisfied |
| 9 | 7 | 96462 | Female | Loyal Cust | 52 | Business t | Business | 2035 | 4 | 3 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 4 | 4 | 0 | satisfied |
| 10 | 8 | 79485 | Female | Loyal Cust | 41 | Business t | Business | 853 | 1 | 2 | 2 | 2 | 4 | 3 | 3 | 1 | 1 | 2 | 1 | 4 | 1 | 2 | 0 | 0 | neutral or dissatisfied |
| 11 | 9 | 65725 | Male | disloyal Cu | 20 | Business t | Eco | 1061 | 3 | 3 | 3 | 4 | 2 | 3 | 3 | 2 | 2 | 3 | 4 | 4 | 3 | 2 | 0 | 0 | neutral or dissatisfied |
| 12 | 10 | 34991 | Female | disloyal Cu | 24 | Business t | Eco | 1182 | 4 | 5 | 5 | 4 | 2 | 5 | 2 | 2 | 3 | 3 | 5 | 3 | 5 | 2 | 0 | 0 | neutral or dissatisfied |
| 13 | 11 | 51412 | Female | Loyal Cust | 12 | Personal T | Eco Plus | 308 | 2 | 4 | 2 | 2 | 1 | 2 | 1 | 1 | 1 | 2 | 5 | 5 | 5 | 1 | 0 | 0 | neutral or dissatisfied |
| 14 | 12 | 98628 | Male | Loyal Cust | 53 | Business t | Eco | 834 | 1 | 4 | 4 | 4 | 1 | 1 | 1 | 1 | 1 | 1 | 3 | 4 | 4 | 1 | 28 | 8 | neutral or dissatisfied |
| 15 | 13 | 83502 | Male | Loyal Cust | 33 | Personal T | Eco | 946 | 4 | 2 | 4 | 3 | 4 | 4 | 4 | 4 | 4 | 5 | 2 | 2 | 2 | 4 | 0 | 0 | satisfied |
| 16 | 14 | 95789 | Female | Loyal Cust | 26 | Personal T | Eco | 453 | 3 | 2 | 3 | 2 | 2 | 3 | 2 | 2 | 4 | 3 | 2 | 2 | 1 | 2 | 43 | 35 | neutral or dissatisfied |
| 17 | 15 | 100580 | Male | disloyal Cu | 13 | Business t | Eco | 486 | 2 | 1 | 2 | 3 | 4 | 2 | 1 | 4 | 2 | 1 | 4 | 1 | 3 | 4 | 1 | 0 | neutral or dissatisfied |
| 18 | 16 | 71142 | Female | Loyal Cust | 26 | Business t | Business | 2123 | 3 | 3 | 3 | 3 | 4 | 4 | 4 | 4 | 5 | 3 | 4 | 5 | 4 | 4 | 49 | 51 | satisfied |
| 19 | 17 | 127461 | Male | Loyal Cust | 41 | Business t | Business | 2075 | 4 | 4 | 2 | 4 | 4 | 4 | 4 | 5 | 5 | 5 | 5 | 3 | 5 | 5 | 0 | 10 | satisfied |
| 20 | 18 | 70354 | Female | Loyal Cust | 45 | Business t | Business | 2486 | 4 | 4 | 4 | 4 | 3 | 4 | 5 | 5 | 5 | 5 | 5 | 3 | 5 | 4 | 7 | 5 | satisfied |
| 21 | 19 | 66246 | Male | Loyal Cust | 38 | Personal T | Eco | 460 | 2 | 3 | 3 | 2 | 5 | 3 | 5 | 5 | 1 | 2 | 4 | 3 | 2 | 5 | 17 | 18 | neutral or dissatisfied |
| 22 | 20 | 39076 | Male | Loyal Cust | 9 | Business t | Eco | 1174 | 2 | 4 | 2 | 4 | 2 | 2 | 1 | 2 | 1 | 5 | 3 | 4 | 3 | 2 | 0 | 4 | neutral or dissatisfied |
| 23 | 21 | 22434 | Female | Loyal Cust | 17 | Personal T | Eco | 208 | 3 | 1 | 3 | 3 | 5 | 3 | 5 | 5 | 2 | 5 | 3 | 3 | 4 | 5 | 0 | 0 | neutral or dissatisfied |
| 24 | 22 | 43510 | Female | Loyal Cust | 43 | Personal T | Eco | 752 | 3 | 5 | 3 | 5 | 5 | 4 | 5 | 3 | 3 | 3 | 5 | 3 | 3 | 4 | 52 | 29 | neutral or dissatisfied |
| 25 | 23 | 114090 | Female | Loyal Cust | 58 | Personal T | Eco | 2139 | 4 | 5 | 4 | 5 | 4 | 3 | 4 | 4 | 4 | 4 | 4 | 2 | 4 | 2 | 0 | 0 | neutral or dissatisfied |
| 26 | 24 | 105420 | Female | disloyal Cu | 23 | Business t | Eco | 452 | 5 | 0 | 5 | 1 | 1 | 5 | 1 | 1 | 4 | 5 | 5 | 3 | 5 | 1 | 54 | 44 | satisfied |
| 27 | 25 | 102956 | Male | Loyal Cust | 57 | Personal T | Eco | 719 | 4 | 4 | 4 | 1 | 5 | 4 | 5 | 5 | 3 | 2 | 4 | 4 | 5 | 5 | 27 | 28 | neutral or dissatisfied |
| 28 | 26 | 18510 | Female | Loyal Cust | 33 | Business t | Business | 1561 | 1 | 1 | 1 | 1 | 1 | 5 | 3 | 4 | 4 | 4 | 3 | 5 | 4 | 2 | 0 | 0 | satisfied |
| 29 | 27 | 14925 | Female | Loyal Cust | 49 | Business t | Eco Plus | 315 | 4 | 4 | 4 | 4 | 2 | 2 | 1 | 4 | 4 | 4 | 4 | 2 | 4 | 2 | 0 | 8 | satisfied |
| 30 | 28 | 118319 | Female | Loyal Cust | 36 | Business t | Business | 3347 | 3 | 1 | 1 | 1 | 1 | 2 | 1 | 3 | 3 | 3 | 3 | 2 | 3 | 2 | 18 | 12 | neutral or dissatisfied |
| 31 | 29 | 75460 | Male | Loyal Cust | 22 | Personal T | Eco | 2342 | 3 | 2 | 3 | 3 | 3 | 3 | 1 | 3 | 2 | 4 | 3 | 4 | 2 | 3 | 19 | 0 | neutral or dissatisfied |
| 32 | 30 | 48492 | Female | Loyal Cust | 31 | Business t | Business | 819 | 4 | 4 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 4 | 3 | 1 | 5 | 5 | 0 | 0 | satisfied |
| 33 | 31 | 27809 | Female | disloyal Cu | 15 | Business t | Eco | 1043 | 2 | 2 | 2 | 3 | 5 | 2 | 5 | 5 | 3 | 1 | 4 | 2 | 4 | 5 | 3 | 0 | neutral or dissatisfied |
| 34 | 32 | 70594 | Female | Loyal Cust | 35 | Business t | Business | 2611 | 4 | 5 | 4 | 4 | 4 | 4 | 4 | 3 | 3 | 4 | 5 | 4 | 3 | 4 | 109 | 120 | satisfied |
| 35 | 33 | 30089 | Female | Loyal Cust | 67 | Personal T | Eco | 1192 | 4 | 5 | 4 | 1 | 2 | 4 | 5 | 5 | 5 | 4 | 3 | 5 | 5 | 5 | 18 | 24 | neutral or dissatisfied |
| 36 | 34 | 58779 | Male | disloyal Cu | 37 | Business t | Business | 1182 | 3 | 3 | 3 | 4 | 1 | 3 | 1 | 1 | 4 | 1 | 3 | 1 | 4 | 1 | 9 | 0 | neutral or dissatisfied |

## Airline Passenger Satisfaction Dataset

# ALGORITHM & DEPLOYMENT:

Before building the dashboard, the dataset was cleaned and transformed to make it suitable for analysis.

Data preprocessing involved checking column data types, handling missing values, and preparing the dataset for visualization.

Two additional columns were created to improve analysis:

- **Delay Category** – to classify flights into On Time, Minor Delay, Moderate Delay, and Severe Delay.
- **Age Group** – to categorize passengers based on age.

These columns help analyze customer behavior more effectively.

Several DAX measures were also created to calculate key metrics such as total passengers, average flight delay, average flight distance, and satisfaction rate.

# ALGORITHM & DEPLOYMENT:

train.csv

File Origin: 1252: Western European (Windows)
Delimiter: Comma
Data Type Detection: Based on first 200 rows

| | id | Gender | Customer Type | Age | Type of Travel | Class | Flight Distance | Inflight wifi service | Departure/Arrival time conve |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 70172 | Male | Loyal Customer | 13 | Personal Travel | Eco Plus | 460 | 3 | |
| 1 | 5047 | Male | disloyal Customer | 25 | Business travel | Business | 235 | 3 | |
| 2 | 110028 | Female | Loyal Customer | 26 | Business travel | Business | 1142 | 2 | |
| 3 | 24026 | Female | Loyal Customer | 25 | Business travel | Business | 562 | 2 | |
| 4 | 119299 | Male | Loyal Customer | 61 | Business travel | Business | 214 | 3 | |
| 5 | 111157 | Female | Loyal Customer | 26 | Personal Travel | Eco | 1180 | 3 | |
| 6 | 82113 | Male | Loyal Customer | 47 | Personal Travel | Eco | 1276 | 2 | |
| 7 | 96462 | Female | Loyal Customer | 52 | Business travel | Business | 2035 | 4 | |
| 8 | 79485 | Female | Loyal Customer | 41 | Business travel | Business | 853 | 1 | |
| 9 | 65725 | Male | disloyal Customer | 20 | Business travel | Eco | 1061 | 3 | |
| 10 | 34991 | Female | disloyal Customer | 24 | Business travel | Eco | 1182 | 4 | |
| 11 | 51412 | Female | Loyal Customer | 12 | Personal Travel | Eco Plus | 308 | 2 | |
| 12 | 98628 | Male | Loyal Customer | 53 | Business travel | Eco | 834 | 1 | |
| 13 | 83502 | Male | Loyal Customer | 33 | Personal Travel | Eco | 946 | 4 | |
| 14 | 95789 | Female | Loyal Customer | 26 | Personal Travel | Eco | 453 | 3 | |
| 15 | 100580 | Male | disloyal Customer | 13 | Business travel | Eco | 486 | 2 | |
| 16 | 71142 | Female | Loyal Customer | 26 | Business travel | Business | 2123 | 3 | |
| 17 | 127461 | Male | Loyal Customer | 41 | Business travel | Business | 2075 | 4 | |
| 18 | 70354 | Female | Loyal Customer | 45 | Business travel | Business | 2486 | 4 | |
| 19 | 66246 | Male | Loyal Customer | 38 | Personal Travel | Eco | 460 | 2 | |

Extract Table Using Examples | Load | Transform Data | Cancel

## Data Cleaning using Power Query

Custom Column

Add a column that is computed from the other columns.

New column name: Custom

Custom column formula

```
= if [Departure Delay in Minutes] = 0 then "On Time"
  else if [Departure Delay in Minutes] <= 30 then "Minor Delay"
  else if [Departure Delay in Minutes] <= 60 then "Moderate Delay"
  else "Severe Delay"
```

Available columns: id, Gender, Customer Type, Age, Type of Travel, Class

<< Insert

Learn about Power Query formulas

No syntax errors have been detected.

OK | Cancel

| Delay Category |
|---|
| Minor Delay |
| Minor Delay |
| On Time |
| Minor Delay |
| On Time |
| On Time |
| Minor Delay |
| Minor Delay |
| On Time |
| On Time |
| On Time |
| On Time |
| Minor Delay |
| On Time |
| Moderate Delay |
| Minor Delay |
| Moderate Delay |
| On Time |
| Minor Delay |
| Minor Delay |
| On Time |
| On Time |
| Moderate Delay |

Custom Column

Add a column that is computed from the other columns.

New column name: Age Group

Custom column formula

```
= if [Age] < 25 then "Young Travelers"
  else if [Age] < 45 then "Adult Travelers"
  else if [Age] < 60 then "Senior Travelers"
  else "Elderly Travelers"
```

Available columns: id, Gender, Customer Type, Age, Type of Travel, Class

<< Insert

Learn about Power Query formulas

No syntax errors have been detected.

OK | Cancel

| Age Group |
|---|
| Young Travelers |
| Adult Travelers |
| Adult Travelers |
| Adult Travelers |
| Elderly Travelers |
| Adult Travelers |
| Senior Travelers |
| Senior Travelers |
| Adult Travelers |
| Young Travelers |
| Young Travelers |
| Young Travelers |
| Senior Travelers |
| Adult Travelers |
| Adult Travelers |
| Young Travelers |
| Adult Travelers |
| Adult Travelers |
| Senior Travelers |
| Adult Travelers |
| Young Travelers |
| Young Travelers |
| Adult Travelers |
| Senior Travelers |
| Young Travelers |

## Feature Engineering – Delay Category & Age Group

# ALGORITHM & DEPLOYMENT:

```
Satisfaction Rate =
DIVIDE(
CALCULATE(COUNTROWS(train), train[satisfaction] = "satisfied"),
COUNTROWS(train)
)
```

Measures the **overall customer satisfaction level**.

```
Average Flight Distance = AVERAGE(train[Flight Distance])
```

Used to understand **typical travel distance patterns**.

```
Average Delay = AVERAGE(train[Departure Delay in Minutes])
```

Helps analyze the **overall delay experienced by passengers**.

```
Total Passengers = COUNTROWS(train)
```

Used to show the **total number of passengers** in KPI cards and charts.

## DAX Measures for Customer Analytics

DAX measures were created to calculate key performance indicators such as total passengers, average delay, average flight distance, and passenger satisfaction rate. These measures help analyze airline passenger behavior and service performance through interactive visualizations.

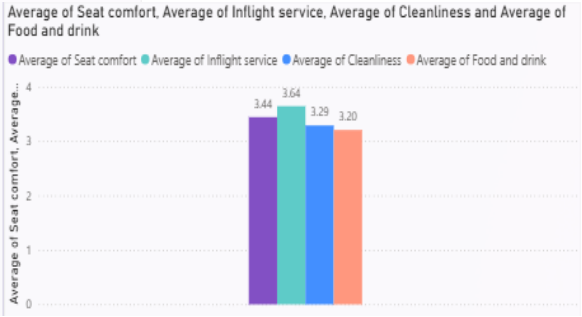
# RESULT:

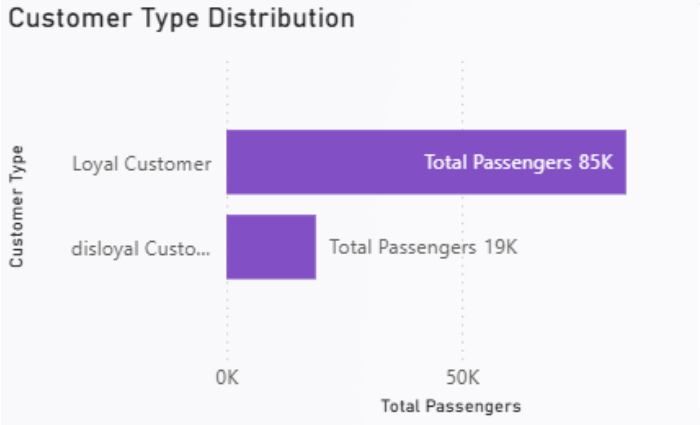
- The Power BI dashboard provides several visualizations that help understand passenger behavior and satisfaction.
- The dashboard includes charts that show the distribution of satisfied and dissatisfied passengers, the difference between loyal and disloyal customers, and how flight delays affect passenger satisfaction.
- It also highlights how service quality factors such as seat comfort and inflight service influence the passenger experience.
- These visual insights make it easier to understand patterns in the data and identify areas where airlines can improve their services.

# RESULT:

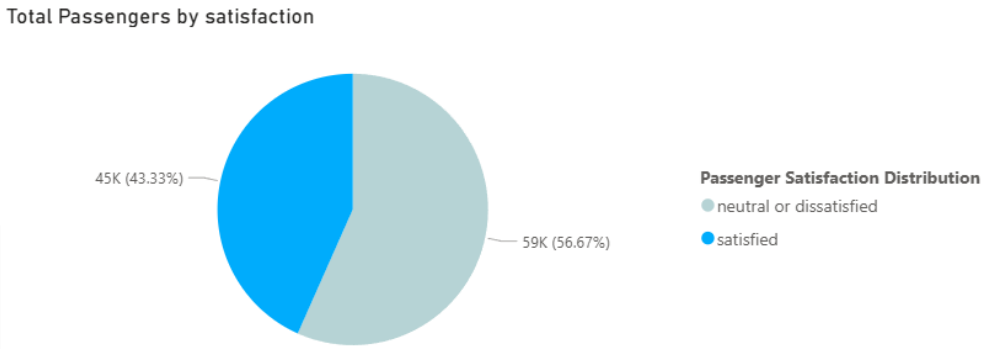
Impact of Flight Delays on Customer Satisfaction
satisfaction
neutral or dissatisfied
satisfied
60K
40K
20K
0K
Total Passengers
27K
32K
13K
18K
5K
5K
On Time
Minor Delay
Moderate Delay
Severe Delay
Delay Category

Passenger Satisfaction by Travel Class
satisfaction
neutral or dissatisfied
satisfied
40K
20K
0K
Total Passengers
32K
11K
7K
29K
5K
Business
Eco
Eco Plus
Class

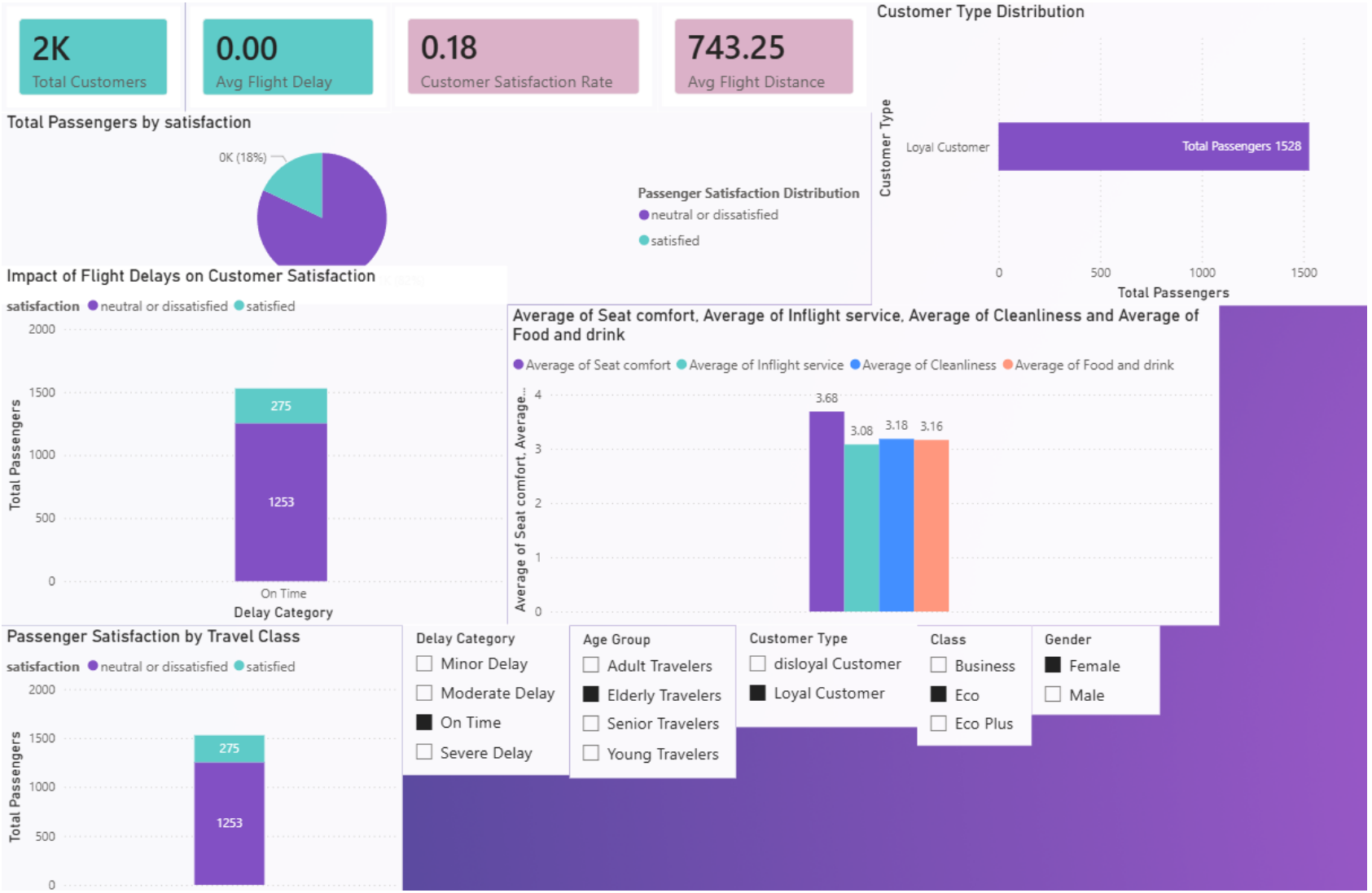

Average of Seat comfort, Average of Inflight service, Average of Cleanliness and Average of Food and drink
Average of Seat comfort
Average of Inflight service
Average of Cleanliness
Average of Food and drink
3.44
3.64
3.29
3.20


Customer Type Distribution
Customer Type
Loyal Customer
Total Passengers 85K
disloyal Custo...
Total Passengers 19K
0K
50K
Total Passengers

45K
Total Customers
12.61
Avg Flight Delay
1.53K
Avg Flight Distance
1.00
Customer Satisfaction Rate


Total Passengers by satisfaction
45K (43.33%)
59K (56.67%)
Passenger Satisfaction Distribution
neutral or dissatisfied
satisfied

# RESULT:




The final Power BI dashboard provides a comprehensive overview of airline passenger behavior and service quality. The dashboard integrates multiple visualizations and key performance indicators to help analyze passenger satisfaction and operational factors affecting customer experience.

The dashboard highlights important metrics such as total passengers, average flight delay, average flight distance, and overall satisfaction rate. It also includes charts that show passenger satisfaction distribution, customer type distribution, and the impact of flight delays on passenger satisfaction.

Interactive filters allow users to explore the dataset by gender, customer type, travel class, age group, and delay category. These filters enable deeper analysis of different passenger segments and help identify patterns in customer behavior.

The dashboard provides valuable insights that can help airline companies understand passenger expectations and improve service quality.

# CONCLUSION:

- This project demonstrates how Power BI can be used to analyze airline passenger data and gain valuable insights into customer behavior and service quality.
- The dashboard highlights important factors that affect passenger satisfaction, such as travel class, service ratings, and flight delays. By visualizing this information clearly, the dashboard helps airlines understand passenger expectations and improve their services.
- Overall, the project shows how data analytics can support better decision-making and enhance the passenger experience

# FUTURE SCOPE:

- In the future, the project can be expanded by integrating real-time airline operational data. Advanced analytics techniques such as machine learning could also be used to predict passenger satisfaction and flight delays.
- Additional datasets could be included to provide deeper insights into airline performance and customer behavior.

LINK FOR THE POWER BI FILE :LINK
LINK FOR THE ppt: LINK

# REFERENCES:


- **Airline Passenger Satisfaction Dataset**
  Kaggle
  https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction
- **Microsoft Power BI Documentation**
  https://learn.microsoft.com/en-us/power-bi/
- **Power BI Data Visualization Guide**
  https://learn.microsoft.com/en-us/power-bi/visuals/power-bi-report-add-visualizations
- **DAX (Data Analysis Expressions) Documentation**
  https://learn.microsoft.com/en-us/dax/

# Thank You